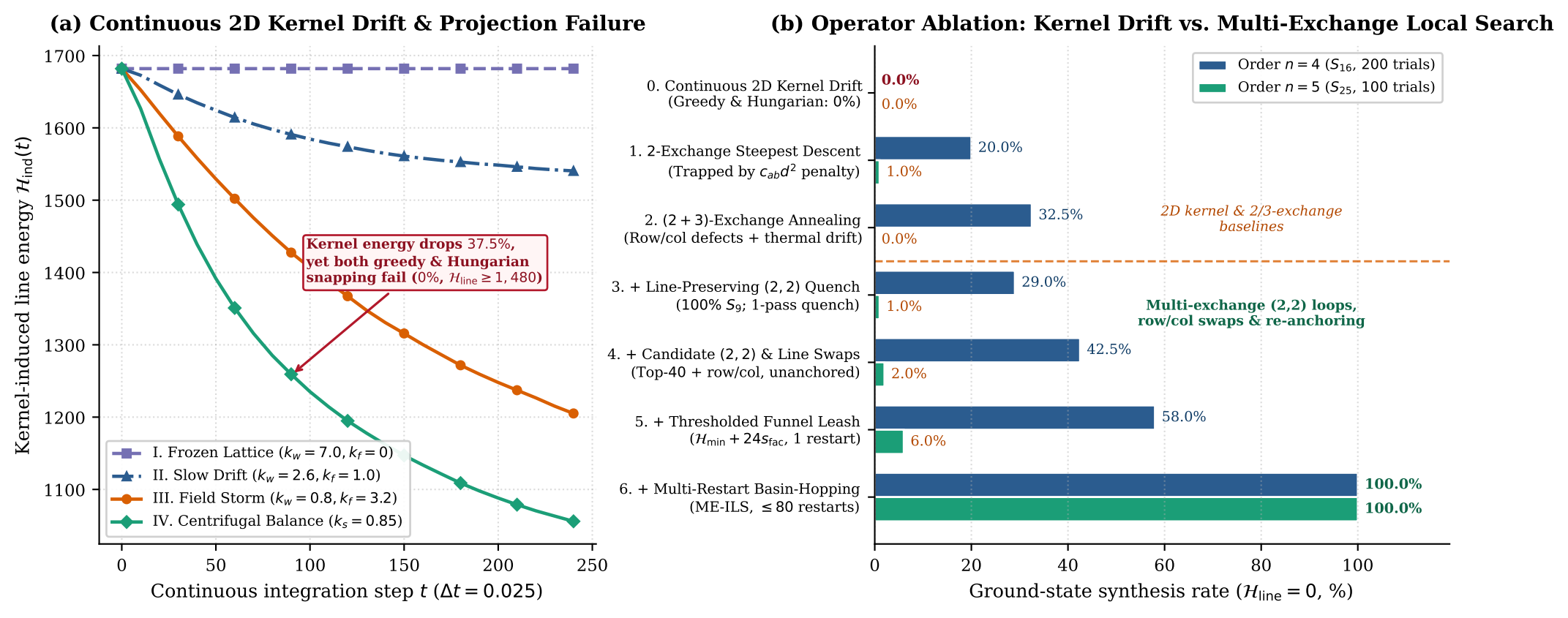

(a) Continuous 2D Kernel Drift & Projection Failure
Kernel-induced line energy $\mathcal{H}_{\text{ind}}(t)$
Continuous integration step t ($\Delta t = 0.025$)
Kernel energy drops 37.5%, yet both greedy & Hungarian snapping fail (0%, $\mathcal{H}_{\text{line}} \geq 1,480$)
I. Frozen Lattice ($k_w = 7.0, k_f = 0$)
II. Slow Drift ($k_w = 2.6, k_f = 1.0$)
III. Field Storm ($k_w = 0.8, k_f = 3.2$)
IV. Centrifugal Balance ($k_s = 0.85$)
(b) Operator Ablation: Kernel Drift vs. Multi-Exchange Local Search
Order $n = 4$ (S_{16}, 200 trials)
Order $n = 5$ (S_{25}, 100 trials)
0. Continuous 2D Kernel Drift (Greedy & Hungarian: 0%)
1. 2-Exchange Steepest Descent (Trapped by $c_{ab}d^2$ penalty)
2. (2 + 3)-Exchange Annealing (Row/col defects + thermal drift)
3. + Line-Preserving (2, 2) Quench (100% S_9; 1-pass quench)
4. + Candidate (2, 2) & Line Swaps (Top-40 + row/col, unanchored)
5. + Thresholded Funnel Leash ($\mathcal{H}_{\min} + 24s_{\text{fac}}$, 1 restart)
6. + Multi-Restart Basin-Hopping (ME-ILS, ≤ 80 restarts)
0.0%
0.0%
20.0%
1.0%
32.5%
0.0%
29.0%
1.0%
42.5%
2.0%
58.0%
6.0%
100.0%
100.0%
2D kernel & 2/3-exchange baselines
Multi-exchange (2,2) loops, row/col swaps & re-anchoring
Ground-state synthesis rate ($\mathcal{H}_{\text{line}} = 0$, %)